

Question 3(a)

We aim to approximate the second derivatives using the **Central Difference Method**. The second derivative of a function $f(x)$ at a point x can be approximated using Taylor expansions.

Consider the function $\phi(x, y)$ evaluated at the point (i, j) , where $\phi_{i,j} = \phi(x_i, y_j)$.

$$\begin{aligned} \text{According to Taylor's theorem: } \phi_{i+1,j} &= \phi(x_i + h, y_j) \\ &= \phi_{i,j} + h \left. \frac{\partial \phi}{\partial x} \right|_{(i,j)} + \frac{h^2}{2} \left. \frac{\partial^2 \phi}{\partial x^2} \right|_{(i,j)} + \frac{h^3}{6} \left. \frac{\partial^3 \phi}{\partial x^3} \right|_{(i,j)} + O(h^4) \\ \phi_{i-1,j} &= \phi(x_i - h, y_j) \\ &= \phi_{i,j} - h \left. \frac{\partial \phi}{\partial x} \right|_{(i,j)} + \frac{h^2}{2} \left. \frac{\partial^2 \phi}{\partial x^2} \right|_{(i,j)} - \frac{h^3}{6} \left. \frac{\partial^3 \phi}{\partial x^3} \right|_{(i,j)} + O(h^4) \end{aligned}$$

Adding equations () and ():

$$\phi_{i+1,j} + \phi_{i-1,j} = 2\phi_{i,j} + h^2 \left. \frac{\partial^2 \phi}{\partial x^2} \right|_{(i,j)} + O(h^4) \quad (1)$$

Rearranging and simplifying, we obtain:

$$\left. \frac{\partial^2 \phi}{\partial x^2} \right|_{(i,j)} = \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{h^2} + O(h^2) \quad (2)$$

Therefore, neglecting higher-order terms, we can approximate:

$$\left. \frac{\partial^2 \phi}{\partial x^2} \right|_{(i,j)} \approx \frac{\phi_{i-1,j} - 2\phi_{i,j} + \phi_{i+1,j}}{h^2} \quad (3)$$

$$\begin{aligned} \text{Similarly, for the second derivative in the } y \text{ direction, we have: } \phi_{i,j+1} &= \phi(x_i, y_j + h) \\ &= \phi_{i,j} + h \left. \frac{\partial \phi}{\partial y} \right|_{(i,j)} + \frac{h^2}{2} \left. \frac{\partial^2 \phi}{\partial y^2} \right|_{(i,j)} + \frac{h^3}{6} \left. \frac{\partial^3 \phi}{\partial y^3} \right|_{(i,j)} + O(h^4) \\ \phi_{i,j-1} &= \phi(x_i, y_j - h) \\ &= \phi_{i,j} - h \left. \frac{\partial \phi}{\partial y} \right|_{(i,j)} + \frac{h^2}{2} \left. \frac{\partial^2 \phi}{\partial y^2} \right|_{(i,j)} - \frac{h^3}{6} \left. \frac{\partial^3 \phi}{\partial y^3} \right|_{(i,j)} + O(h^4) \end{aligned}$$

Adding equations () and ():

$$\phi_{i,j+1} + \phi_{i,j-1} = 2\phi_{i,j} + h^2 \left. \frac{\partial^2 \phi}{\partial y^2} \right|_{(i,j)} + O(h^4) \quad (4)$$

Simplifying, we get:

$$\left. \frac{\partial^2 \phi}{\partial y^2} \right|_{(i,j)} \approx \frac{\phi_{i,j-1} - 2\phi_{i,j} + \phi_{i,j+1}}{h^2} \quad (5)$$

Question 3(b)

Laplace's equation is:

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \quad (6)$$

Substituting the approximate expressions derived in part (a):

$$\frac{\phi_{i-1,j} - 2\phi_{i,j} + \phi_{i+1,j}}{h^2} + \frac{\phi_{i,j-1} - 2\phi_{i,j} + \phi_{i,j+1}}{h^2} = 0 \quad (7)$$

Multiplying both sides by h^2 :

$$(\phi_{i-1,j} - 2\phi_{i,j} + \phi_{i+1,j}) + (\phi_{i,j-1} - 2\phi_{i,j} + \phi_{i,j+1}) = 0 \quad (8)$$

Combining like terms:

$$\phi_{i-1,j} + \phi_{i+1,j} + \phi_{i,j-1} + \phi_{i,j+1} - 4\phi_{i,j} = 0 \quad (9)$$

Rearranging:

$$4\phi_{i,j} = \phi_{i-1,j} + \phi_{i+1,j} + \phi_{i,j-1} + \phi_{i,j+1} \quad (10)$$

Finally, dividing both sides by 4:

$$\phi_{i,j} = \frac{1}{4} (\phi_{i-1,j} + \phi_{i+1,j} + \phi_{i,j-1} + \phi_{i,j+1}) \quad (11)$$

where $1 \leq i < n$ and $1 \leq j < n$, as this formula applies only to *interior points*. The values at the boundary points are specified by the problem's boundary conditions.